NAME : Simran Chouhan **SEMESTER:**4th

SECTION:24_AIT_KRG-G1 **UID:**24BAI70124

SUBJECT CODE:24CSH-298

FACULTY'S NAME: Mr .SHALABH BHATIA

Topic : Experiment 4

AIM:

To design and implement PL/SQL programs utilizing conditional control statements such as IF–ELSE, ELSIF, ELSIF ladder, and CASE constructs in order to control the flow of execution based on logical conditions and to analyze decision-making capabilities in PL/SQL blocks.

Software Requirements:

- **Database Management System:**
  - PostgreSQL
- **Database Administration Tool:**
  - pgAdmin

Objective:

Implement control structures in PL/SQL (IF-ELSE, ELSE-IF, ELSE-IF LADDER, CASE STATEMENTS in PL-SQL BLOCK)

Problem Statement

Develop and execute PL/SQL programs that demonstrate the use of **conditional control statements**. The programs should employ **IF–ELSE**, **ELSIF**, **ELSIF ladder**, and **CASE** statements to evaluate given conditions and control the flow of execution accordingly, thereby illustrating decision-making capabilities in PL/SQL blocks.

TABLE CREATION:

```
CREATE TABLE student_lab (
    student_id SERIAL PRIMARY KEY,
    student_name VARCHAR(50),
    marks INTEGER,
    day_no INTEGER
);


INSERT INTO student_lab (student_name, marks, day_no) VALUES
('Manan', 95, 1),
('Aarav', 78, 3),
('Simran', 62, 5),
('Kunal', 45, 7),
('Neha', -5, 9);
```

1. Problem Statement – IF–ELSE Statement

Write a PL/SQL program to check whether a given number is positive or non-positive using the **IF–ELSE** conditional control statement and display an appropriate message.

CODE:

```
DO $$

DECLARE

   r RECORD;

BEGIN

   FOR r IN SELECT * FROM student_lab LOOP

      IF r.marks > 0 THEN

         RAISE NOTICE '% has POSITIVE marks (%).', r.student_name,
r.marks;

      ELSE

         RAISE NOTICE '% has NON-POSITIVE marks (%).',
r.student_name, r.marks;

      END IF;

   END LOOP;

END $$;
```

2. Problem Statement – IF–ELSIF–ELSE Statement

Write a PL/SQL program to evaluate the grade of a student based on the obtained marks using the **IF–ELSIF–ELSE** statement and display the corresponding grade.

CODE:

```
DO $$

DECLARE

   r RECORD;

BEGIN

   FOR r IN SELECT * FROM student_lab LOOP

      IF r.marks >= 90 THEN

         RAISE NOTICE '% : Grade A', r.student_name;

      ELSIF r.marks >= 75 THEN

         RAISE NOTICE '% : Grade B', r.student_name;

      ELSIF r.marks >= 60 THEN

         RAISE NOTICE '% : Grade C', r.student_name;

      ELSE

         RAISE NOTICE '% : Grade F', r.student_name;

      END IF;

   END LOOP;

END $$;
```

3. Problem Statement – ELSIF Ladder

Write a PL/SQL program to determine the performance status of a student based on marks using an **ELSIF ladder** and display the appropriate result.

```
DO $$
```

```
DECLARE
   r RECORD;
BEGIN
   FOR r IN SELECT * FROM student_lab LOOP
      IF r.marks >= 90 THEN
         RAISE NOTICE '% : EXCELLENT', r.student_name;
      ELSIF r.marks >= 75 THEN
         RAISE NOTICE '% : VERY GOOD', r.student_name;
      ELSIF r.marks >= 60 THEN
         RAISE NOTICE '% : GOOD', r.student_name;
      ELSIF r.marks >= 50 THEN
         RAISE NOTICE '% : AVERAGE', r.student_name;
      ELSE
         RAISE NOTICE '% : POOR', r.student_name;
      END IF;
   END LOOP;
END $$;
```

4. Problem Statement – CASE Statement

Write a PL/SQL program to display the name of the day based on a given day number using the **CASE** conditional statement.

CODE:

```
DO $$
DECLARE
    r RECORD;
    day_name TEXT;
BEGIN
    FOR r IN SELECT * FROM student_lab LOOP
        day_name := CASE r.day_no
            WHEN 1 THEN 'Monday'
            WHEN 2 THEN 'Tuesday'
            WHEN 3 THEN 'Wednesday'
            WHEN 4 THEN 'Thursday'
            WHEN 5 THEN 'Friday'
            WHEN 6 THEN 'Saturday'
            WHEN 7 THEN 'Sunday'
            ELSE 'INVALID DAY'
        END;

        RAISE NOTICE '% : Day % is %', r.student_name, r.day_no, day_name;
    END LOOP;
END $$;
```

OUTPUT:

Data Output Messages Notifications

	student_id [PK] integer	student_name character varying (50)	marks integer	day_no integer
1	1	Manan	95	1
2	2	Aarav	78	3
3	3	Simran	62	5
4	4	Kunal	45	7
5	5	Neha	-5	9

Total rows: 5 Query complete 00:00:00.112

Data Output Messages Notifications

```
NOTICE:  Manan : Grade A
NOTICE:  Aarav : Grade B
NOTICE:  Simran : Grade C
NOTICE:  Kunal : Grade F
NOTICE:  Neha : Grade F
DO

Query returned successfully in 32 msec.
```

Total rows: Query complete 00:00:00.032

Data Output Messages Notifications

```
NOTICE:  Manan has POSITIVE marks (95).
NOTICE:  Aarav has POSITIVE marks (78).
NOTICE:  Simran has POSITIVE marks (62).
NOTICE:  Kunal has POSITIVE marks (45).
NOTICE:  Neha has NON-POSITIVE marks (-5).
DO

Query returned successfully in 43 msec.
```

Total rows: Query complete 00:00:00.043

```
Data Output   Messages   Notifications

NOTICE:  Manan : EXCELLENT
NOTICE:  Aarav : VERY GOOD
NOTICE:  Simran : GOOD
NOTICE:  Kunal : POOR
NOTICE:  Neha : POOR
DO

Query returned successfully in 32 msec.

Total rows:   Query complete 00:00:00.032
```

```
Data Output   Messages   Notifications

NOTICE:  Manan : Day 1 is Monday
NOTICE:  Aarav : Day 3 is Wednesday
NOTICE:  Simran : Day 5 is Friday
NOTICE:  Kunal : Day 7 is Sunday
NOTICE:  Neha : Day 9 is INVALID DAY
DO

Query returned successfully in 32 msec.

Total rows:   Query complete 00:00:00.032
```

LEARNING OUTCOMES:

- Ability to design and use a single relational table effectively to support multiple logical conditions and decision-making scenarios.

- Ability to implement conditional control structures (IF–ELSE, IF–ELSIF–ELSE, and CASE) using PL/pgSQL for data-driven decision making.

- Ability to retrieve and process table records using loops and apply logical branching based on attribute values.

- Improved understanding of how database control structures enhance data validation, classification, and interpretation within stored program blocks.